

Optimal control of the impacts of COVID-19 via non-pharmaceutical interventions

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In collaboration with Zaid Harchaoui
and Adrian Raftery

CSDE Lightning Talks, Winter 2022

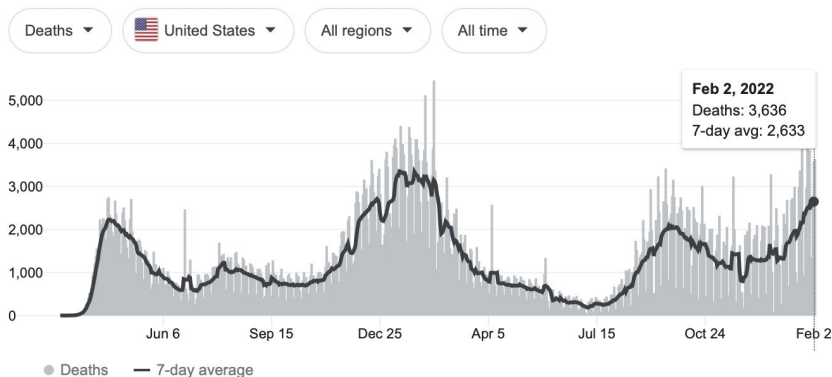
UNIVERSITY *of* WASHINGTON



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Impacts of COVID-19



Each day shows deaths reported since the previous day · [About this data](#)

🌐 Cases

From [The New York Times](#) and [Our World in Data](#) · Last updated: 1 day ago



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Learning loss due to school closures during the COVID-19 pandemic

Per Engzell, Arun Frey, and Mark D. Verhagen

Viewpoint

October 12, 2020

The COVID-19 Pandemic and the \$16 Trillion Virus

David M. Cutler, PhD¹; Lawrence H. Summers, PhD²

» [Author Affiliations](#) | [Article Information](#)

JAMA. 2020;324(15):1495-1496. doi:10.1001/jama.2020.19759



Decision-making under uncertainty

The pandemic has posed significant challenges to decision-makers at every level of government

Difficult real-time decisions with limited data

Contentious debate about non-pharmaceutical intervention (NPI) policies that reduce viral transmission but present severe economic and social consequences:

- Non-essential business closures
- School closures
- Stay-at-home orders
- Social distancing measures

Goals and contributions

We develop a statistical decision framework based in optimal control theory to address the economic and public health trade-offs of NPIs.

- Derive optimal NPI policies
- Quantitatively assess the efficacy of policies that have been implemented during the pandemic

A number of gaps in the optimal control literature modeling NPI mitigation of COVID-19 severely limit its value in informing and evaluating policies

- We model multiple specific NPIs calibrated to US state-level COVID data.
 - Prior work models a catch-all “lockdown” NPI
- We account for uncertainty in the relevant model parameters to derive robust policies
 - Prior work ignores parameter uncertainty, which is problematic.



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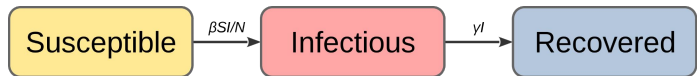
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Goals and Contributions

Develop a statistical decision framework based in optimal control theory to address the economic and public health trade-offs of NPIs.

- Model multiple specific NPIs
- Account for and quantify parameter uncertainty

Epidemiological SIR Model

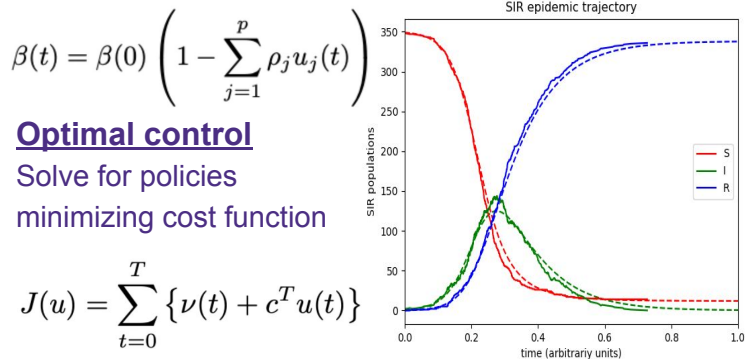


$$s(t+1) - s(t) = -\beta(t)s(t)i(t)$$

$$i(t+1) - i(t) = \beta(t)s(t)i(t) - \gamma i(t)$$

$$r(t+1) - r(t) = \gamma i(t)$$

Controlling transmission with NPIs



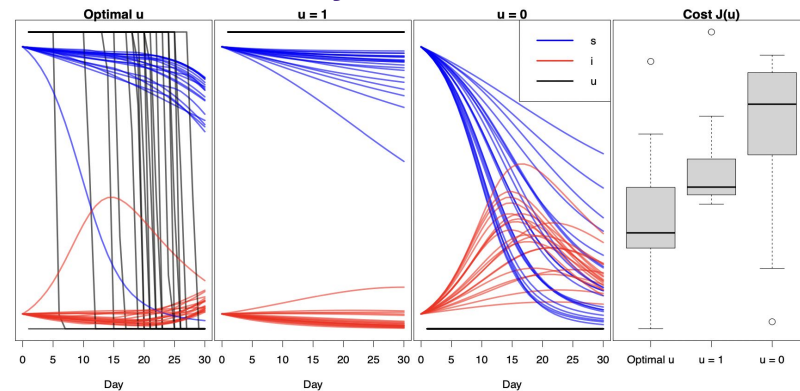
$$\beta(t) = \beta(0) \left(1 - \sum_{j=1}^p \rho_j u_j(t) \right)$$

Optimal control

Solve for policies minimizing cost function

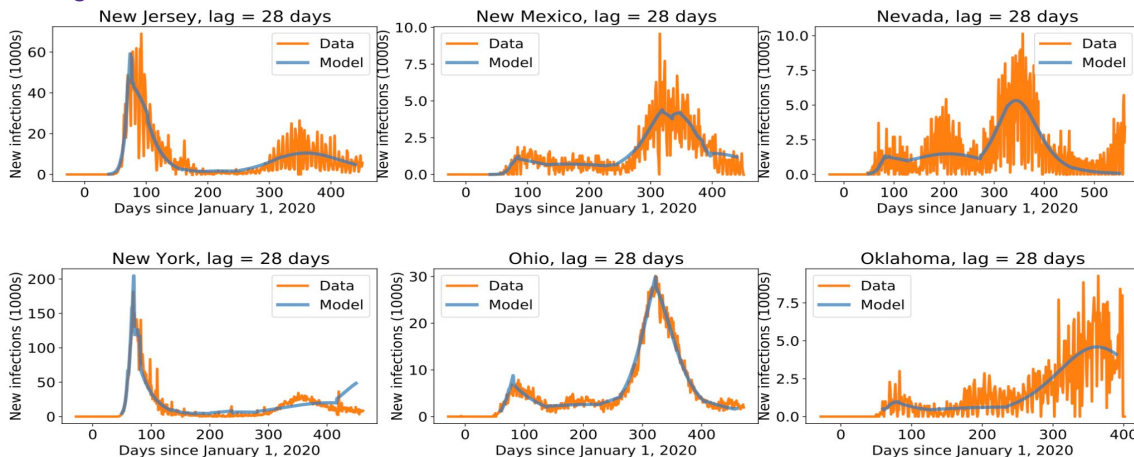
$$J(u) = \sum_{t=0}^T \{ \nu(t) + c^T u(t) \}$$

Parameter Uncertainty



Results

Fitting the control model to state-level COVID data



Data

State-level NPI policies:
Oxford COVID-19
Government Response
Tracker



State-level COVID
transmission data: JHU
Center for Systems
Science and Engineering



Center for Systems Science
and Engineering